

[illegible]

LGDFFRKSKEKIGKEFKRIVQRIKDFLRNLPVPTES
CCGGIKKEIEAIIKKEQEAIIKKLIQLIVWGIKQIQARIL
KNSQGNKENICFKDADCPQDICSYPFKPKCNIYGYCSC
AILECREDSHCVTIKICVLPRKPECRNNACTCYKGGFSFHH
ATCDLASGFGVSSSLCAAHCLVKGYRGGYCKNKICHCRDKF
DDGIEMTEEEVKRGILDLVTHVAKNLAAQLLDKLCCKMTGC
ELRPEDDMKPGSFDRSIPENNIMRTIIEFLSFLHLKEAGAL
FDNPFPGCPADEGKCFDHCNNKAYDIGYCGGSYRATCVCYRK
IKKEIEAIIKKEQEAIIKKIEAIEKLLQLTVWGIAQLQARIL
GALWGAPAGGVGALPGAFVGAHVGAIAAGGFACMGGMIGNKFN
KNIGNSVSLRNKGVCMPGKCAPKMKQIGTCGMPQVKCKRK
MLLAIAFLASVCVSSKRKRAVKRVGRRLKKLARKIARLGVA
RPVSCDKHYDCRRKVKIVGCIFFQEKPMCINSMCTCIREIVP
SGISGPLSCGRNGGVCIPIRCPVPMRQIGTCFGRPVKCCRSW
YKQCHKKGHCFCPEKICIPPSSDFGKMDCRWRWKCKKGS
AAAAGSVWGAVNYTSDCNCECKRRGYKGGYCGSFANVNCWCET
DRCTKRYGRCKRDCLESEKQIDICSLPRKICCTEKL YEEDDMF
KYYGNGLSCNKGCSVDWSKAISIIGNNAVANLTTGGAAGWKS
YKQCHKKGHCFCPEKICIPPSSDFGKMDCRWRWKCKKRS
AIHRALISKRMEGHCEAECLTFEVKIGGCRAELAPFCKNRKKH
AKVNCDDADCLEVLCVFGSKAECVVNICICVPPRFKGFDEHFR
CGGNLLRAIEAQHLLQLTVWGIKQLQARILAVERYLKDQGGC
DKDNKRFBALLGDFFRKSKEKIGKEFKRIVQRIKDFLRNLPVPTES
LQHYREVAAKSSENDRLRLLLKQMCPSLDVDSIIPRTPDVLHED
QEIENGHIHPCKKNEDCNHMCVMPGLPWCHENNLFCYENAYGNTR
SSLLEKGLDGAKKAVGGLGKLGKDAVEDLESVGKGAHVHDVKDVL
KSCCPNTTGRNIYNTCRFAGGSRERCAKLSGCKIISASTCPSDYPK
MHIETVTTCIYSDCPEDMCYPPKKSFCSTFEILSIERKVGVCCEI
VKSLLLIKVRSFIPCQRSDDCPRNL CVDQIIPTCVWAKCKCKNYND
EIDADCPQICMPPEYVRCVNHRCGWVNTDSDLTQEFTRSKQYIIS
INSDGYLECTTDYDCREEWLCPPDMEAKCFVSFALARFLSKGKCLCV
KVCQRQSAQFKGPCVSDKNCAQVCLQEQQWQONCDQPFRRCKCIRQC
RKGPPGGRTYIPICISDDDCIVAQPPYVLLCVNNFCTYFKDDDLQR
AIIYLSLFLFVFEKRECDTNFDCQKQFSTQAECLLWCIRGYCMSIPN
ATYYNGVYCNKQKQWVDSRARSEIIDRGVKAYVNGFTKVLGGIGGR
LNGNRHGKDRCFKSDCPKYMCPSSLVAKCIKKLCSCKRPGQLIQLNPK
MTQFLGALLLTGVLGYIPYKYLTMIGLVSEKKQGYQYSCIIDFLLKHV
ELCEKASKTWSGNCGNTKHCDNQCKSWEGAKHGACHKRSKGHKCFCYFNC
MKVFFLFAVLFCVLQTNVHISHQEARGPSFKICVGLGPRWARGCSTGN
MNDNSOGLQANPQYTIHYLSQEITRLTQENAMLKAYIQEQNEKSKSAEEE
NLCEKASKTWSGNCGNTKHCDNQCRNWEAGAEHGACHVRNGKHKCFCYFNC
NLCEKASKTWSGNCGNTKHCDTQCRNWEAGAKHGACHKRNKGWKCFCYFNC
NLCEKASKTWSGNCGNTKHCDTQCRSWEGAKHGACHVRSGKHKCFCYFNC
NLCEKASKTWTGNCGNTKHCDNQCRNWEAGAAHGACHKRNKGWKCFCYFNC
NLCEKASKTWTGNCGNTKHCDNQCRSWEGAAHGACHVRNGKHKCFCYFNC
NLCEKASKTWTGNCGNTKHCDTQCKNWEAGAKHGACHVRSGKHKCFCYFNC
NLCEKASKTWTGNCGNTKHCDTQCKSWESAKHGACHVRSGKHKCFCYFNC
LDRPKKPGLCPPRPQKPPCVRECKNDWRCPGERKCCRYGCIYECRDPIFVK
LVNKINKFNQYVARFIPYIATKIPIKKNIAPMIKQKSDAELLKNEVFRVRK
QARQLLSGIVQQNNLLRAIEAQHALQATVWGIKQLQARILAVERYLKDQ
QARQLLSGIVQQNNLLRAIEAQHLLQLTVWGIKQLQARILAVERYLKDQ
QKLCERPSRTWSGVCNNSACKNQCNLEKARHGSCNYRFP AHKCICYFPC
QKLCQRPSTWSGVCNNSACKNQCNLEKARHGSCNYRFP AHKCICYFPC
VTCDLLSFGGVVGDSSACANCLSMGKAGGSCNGGICECRKTTFKELWDQRF
VYDSKYFQIASPCVNDKDCPQFKNNNVRRCRRGFCVNSGGATQKCLGCPSLK
MAMYEVKKSYPDLEKGQYLKSGKRVEMTVKRAEYVNNKKLKEHGVILERVKEE
MKITNCKIKRETIVYEVLTSGNQPFYELPKDLSSHNARKYLEFISQKIRWR
PGDIPPGIRNTVCLMQQGHCRFLMCRSGERKGDICSDPWNRCCVPYSVKDRR
QVVRPGSCPNDVPIPLGLCRTTCQTDANCQEGRKCCCKNGCGFMT CETARF
RMKQIEDKIEEIESKQKKIENEIARIKKLLQLTVWGIKQLQARILAVERYLK
DGVKLCEKASKTWSGNCGNTKHCDKQCRSWEKAAHGACHVRSGKWKCFCYFNC
DGVKLCEKPSKTWSGHCNNTKHCDTQCRNWEKAKHGACHVRNGKWKCFCYFNC
DGVKLCEKPSQTWTGHCNNTKHCDKQCKNWEKAKHGACHVRNGKWKCFCYFNC
DGVKLCEKASKTWSGHCNNTKHCDNQCRSWEGAKHGACHVRNGKWKCFCYFNC
DGVKLCEKASKTWTGNCNNTKHCDKQCKNWEKAKHGACHVRNGKWKCFCYFNC
DQMSDGVNYGKSSLSKGGAKCGLGIVGGLATIPSGPLGWLAGAAGVINSCMK
MAGFLKVVQLLAKYGSKAVQAWANKGKILDWLNAGQAIDWVVS KIKQILGIK
DCSQDCAACSILARPAELNTETCILECEGKLSSNDEGGLCKEFLHPSKVDLPR
KLTFIQSTAAGDLYNTNTHKYVYQQTQNAFGAAANTIVNGWMGGAAGGFLHH

LRDAEQKLKNSNOATTAQLKRASDAVQKQSAKHKALVEQYKQEGNQVQKTKSAK
MFVYDLILFISLILVVTGINAEADTSCHSFDDCPWVAHHYRECIEGLCAYRILY
MSLEEIKLWLRIDYNFENDLIEGLIQSAKSELLLSGVPDYDKDDLEYPLFLYS
SIGNKGISFETCTAIEGLCFFGCKLGWVWIAYCNNIMSCCRKDTDFVLPQTKGI
ATTGCSQPQCIIIFDPICASSYKNGRRGFSSGCHMRCYNRCHGTDYFQISKGSKCI
MKIIVVLAVLMLVSAQVCLVSAAEEMGHSSDNELSSRDLVKRFFLPFCAYKGTCH
MSYTKRALEARGYDFDTMSMLEKMNALLELEEPEFQEQMRKEYEEFSKECATNGE
MVNRSVAFSAFVLILFVLAIASVSGFCFVPGKTCATDRECGSCHCNNWKGKCN
DDDDEGGGGGGGGGGGGERRRPRPPYLPRPRPPFFPPRLPPRIPPGFPPRFP
MEMMNAEKHMOMMOMLONCVIDKYVSHDEYEELIAIDKHGKMFIFYPNTEDDTNE
MFTLKKSLLLFFLGTINLSLCEQERNAEEERRDDLGERQAEVEKRFFPIVGKLLSG
MTNYIVIFFLALFLIVIDVSAILECIFDIDCPTKKCAPPLVAKCDMYECYCRCPNN
PGYRWMCLRRFIIFLFIILLCLIFLLVLLDYQGMPLVCPCLIPGSSTSTGPCRTCMTT
PIRLSKEKINDLLQRSQGLTSSQHEIVHFTDVFIAAGSGPISCTYARHIIDNTSTTK
TRAKVHKFIYIMIVFLSIFHVVNSYVVMCEKSDCVDSFCVPPNPVKCRVVCKLPK
AEIFKFVYKWLFLVSLFLVIVAKEDDIECVTDADCYEKLPAQRAVMKCIQGFCKIHI
GWWNAFKSIGKLLKSKLAKDITKMAQRAKEYVVKLNGPPEEEVAIDALMNSLDY
MAKILMFVYALIIFISLVITGRSTINVMCYDHDPCFVLDHIAECKGGVCEYTAFFYE
MAQILISVHALSVFIFPFLVVIIRDKPAIPCKFHADCPIMLSIVVECINNVEFIYI
MTQISISFYALIIFFSLFLVVTNGRNKTCNYSSECLFHNCPLGWMMCKFTYFCACSR
AQAEPLQARADEAAAQEQPGADDQEMAHAFTHWESAALPLSDSARGLRICICGRGICRL
ELRPEDDMKPGSFDRSIPENNIMRTIIEFLSFLHLKEAGALDRLLDLPAASSEDIER
FSRYARMRDSRPWSDRKNYSQPGQFTYPEKAPPEKLIKWNNEGSPIFEMPAEGGHIEP
MAEFLHMTYVMIIIFIFLFLSLIDAENVHRCIEYTDPCEDMCHLPLVVVCHDHICKCLRLP
MTQILLFVHVLIISFLSLLLIVTNSVEIPETPCESDAECPPYSPSLYARCIDGCTFLS
GPMRIPEKHRIVREYIRKFGQLNEFVQETENAWYYIKNIRKKVHEVKKDPGLLKYPVKP
LVATGMAAGVAKTIVNAVSAGMDIATASLFSGAFTAAGGIMALIKKYAQKKLWKQLIAA
MAAIKFIYTMFLFIFLFVVPKVDALAGCITDADCVIKKSSSSCRIKCIDFRCLCPTGF
MKITNCKIKKETIVYEVLTSGNQPFYELPKDLSSHARKYLEFISQKIDGDKLTKEDSL
MSQILKFFFATVLIFALFLSATNGKFNCEYEDTDCPIQMCGYPFNVDCVGNKCTCVYNP
MVQTPKLVYVIVLLLSIFLGMTICNSSFSHFFEGACKSDKDCPKLHRSNVRCRKGQCVQI
RRLRPRRPRRPRRPRRPRRSLPLRPQPRRIPRPILLPWRPPRPIPRPQPPIPRWL
MAGIIFVHVLIIFLSLFLHVVKNDDGSFCFKDSDCPDEMCPSPLEKMCYFLQCKCGVDTIA
MCGIVGIAGVMPVNOQSIYDALTVLQHRGQDAAGIITIDANNCFRLRKANALVSDVFEARM
MEDIVKFVYVIIIFLSIFIIATNMEAKTICIGSDCRNERCMPGKIPVCSEGWDCIGFIP
MNMNFTKLFVAVLLAALVLLGQTEAGGLKKLGKKEGAGKRVFKASEKALPVVVGKAIK
QVTPVMKWCWGSGRCRTTCKESEVYYILCKTEAKCCVDPKYVPVKPKLTDNTSLESTSAV
MAETLKFFVYLILFISILFVIVCDSSYIPIHPCTTVKDCPEVKNYKSRLKGLCISGRRL
MKFTATFLVLMVVLMAEPGEGYWRFRNHRGERLSQRHFADVEQELDKRSVDDEPSSIAFD
MQRRKNMANNHMLIYAMIICLFPYLVTFTAITCDCNEDCLNFFTPLDNLKCIDNVCEVFM
MVEIQKL VYVLIIFLSIFLEMIVSNCTFIGFQDNPCKTNDNCRKVRGVNLRCRNGHCVMILO
NRWYCNSAAGVGGAAGCVLAGYVGEAKENIAGEVRKGWGMAGGFTHNKACKSFPGSGWASG
MAEIKFVYIILCVSLLLIGEASGKECVTDADCNLYYGNKWPLICSNIGYCLSSYEPPHK
MAEVITTVLEMPNTVIATAVGDKIMEADAYRHYLESINVDPRFVLEEYIAMVAEKDQEVWKHV
MFTLKKSLLVLFGLGIVSLSVCEHNDADEEDGGEVTEGEEVKRAALKGCWTNSIPPKPCSGR
MPEQLSVRKVRLVRDLKQVEFADILGVNAKTGVHWEKDTNLSNVTVYALAKLYDIEVDQIKV
MQIGKNMVETQKLVYVILLFLSIFLFTNSPLSQIIFSECKTDKDCPKYQRANIRCRKGQCVRI
MTHIFKFVYALIIIFLSIYVAVNDCIRIHCKDDFDCIENRLQVGCRLQREKPRCVNLVCRCLRR
MTLTVQARQLLSGIVQQNNLLRAIEAQHLLQLTVWGKQLQARILAVERYLKDQQLLGIWG
GGVKGEEKRVCPPDYVRCIRQDDPQCYSDNDCGQDQECFQWQCGFKCVLPVKDNSEEQIPQSKV
MAKTLNFVCAMILFISLFLVSKNVALYIECKTDADCPISKLNMYNWRICKSSCHLYKVIQFMV
MAQKFMFFYALIIIFLSSFYVIINTIDPPHHITNHEIPCKYNHDCPTILDYISICPYHYCEFWRTY
MGHLPDDVGSWATHCINLVCDGELLEDKKSQTYTCEDCGEVVTLDLNEHLEKEAESYLKFFGYK
MRYSLSVSILAILLEFIYIVVLFSPCLLVTDAYNITCNSALDCASNRCILPGMPICVTNKCLCV
MWITMTIVFAILLVCISINSDRAREIQALRYMNDYLLDEVVKTGKYNGLKEYRIELKRMNNDIKK
RICSRDKNCVSRPGVGSIIGRPGGGSIGRPGGGSVIGRPGGGSPPGGGSFNDEFIRDHSDGNRFA
VRDGYIADDDKNCAFCGRNAYCDDECKKNGAESGYCQWAGVYGNACWCYKLPDKVPIRVPGKCNNG
MDEILKFVYTLIIFFSLFFAANNVDANIMNCQSTFDCPRDMCSHIRDVICIFKKCKCAGGRYMPQVP
MLTIACIVLIVFCGYLFLVKCLASLTCFALFVLPGAEEKVAAVVGVLVWNVFAAVVWTAINFLFHVA
MNKILKFVYEMILFSLFHLAREVHDVAHTDIPCEPDADCPKSLHEYFEMKCIDKKCEWSRKTSLIP
MASLKKSLFLVFLFGFVSLSCIEEEKREDKEDEGENEEAENHEERSEEKRLPLLASLVGGLLGKRS
MKVLLLFVAVFLVQRNSGDIPPGIRNTVCFMQRGHRLFMCRSGERKGDICSDPWNRCVSSSIKNR
MTQYLVTTFKDSTGRKHITHITAKSNQRFTVVEAESKEEAKKEYEKQVVRDAVIKVGQLFENIRECGK
GFFGNTWKKIKGKADKIMLKAVKIMVKKEGISKEEAQAKVDAMSKKQIRLYVLKYYGKKALQKASEKL
GFFGNTWKKIKGKTDKIMLKAVKIMVKKEGISKEEAQAKVDAMSKKQIRLYLLKHYGKKALQKASEKL
MFTLKKSMLLLFFLGTISLSLQDEGADEDHGGEMTTEEKRLMDIFKVAVNKLLAAGMKNPRCKAAHC
MFTLKKTLLLLFFLGTINLSLCKEERDADEERRDDPKRDVEVEKRLSGILKLAFKIPSVLCAVLKNC
MFTMKKSLLLLFFLGTINLSLQEEERNADEEERRDERDVEVEKRVIPFVASVAAEMMHVYCAASKKC
MTEILKFVCMIIIFISSFIVSKSLNGGGKDKCFRSDCPKHMCPSSLVAKCINRLCRCRRPELQVQLNP

DRLRQKVAIQRTLRLGLAVSRKVGNAVVRNRIKRRLEAFRQQSVRTDVLVVARPSARQLSMRAMGAYLQ
MAKIVYFVYSMIIFLSLFLVTTKAAERIYRCLDHSCHPTFMCSPGLKPKCMNPKVCKCVPVQSRKYALY
YRIKKDSDFQRIYRLGISVSKKLGNVLRNLIKRAIREAYRLNIDEKIDIIVIAVSSKDIDKQIQNSLE
MATILMYVYITILFISILTVLTEGLYEPLYNFRDPDCRRNIDCPSYLCVAPKVPRCIMFECHCKDIPSDH
MFTLKKSLLLLFLLGTINLSLCEQERDADEEERRDDPDERDVEVEKRFPLLAGLAANFLPKLFCKITRKG
NRMRRSADFETTVRVGLIIAKSVGSAVERHRVARRLRHVAGSIVKELDHVVIRALPSSRHVSSARLEQQLR
MAIKFIYTMFLFILLFVVPTKVDGRITHDPSTRSTVSGGFGKCVRDADCVDEVCSGPCNKRKCVGFECQCPL
MEIKYLLTVFLVLLIVSDHCQAFSLIPSAISGLISAFKGRKRDLNGQIDHFKNFRKRDAELEELLKSLPIY
SDDPKESEGDHLVCVKTTSVLRPGHITNLELIKAGGHCPTANLIATKKNRKLCLDLQAALYKKKIKKLLES
ARLHRPSEFAAALRLGLVIAKRAAARAVTRNTLKRVIREFRARRALDYVVRHLHSLTPASLTALKRSARAEDV
MFTMKKSLLLLFLLGTISLSLCEEERGADEDDGVELTEEEVKRGLLSFKGVAKGVAKDLAGKLEKLCKITGC
MQRLDNMAKNVKFIYVILLFIFLVIIVCDSAFVPSNGPCTTDDCKQVKGYIARCRKGYCMQSVKRTWSSYSR
MFSKKSLLVLFFLGTISLSLCEEERGADEDAVEFTEEEKRSILSTLKDVGISALKNAGSGVLKTLCKLNKNCEK
MLSTLNEIKLGQKTQEQVNIKLDKTLDAIQKEREIDEKNKENDKNIRDMMKMWVLGLVGTIFGSLIALLRMLMGI
MLTLKKSMLLLFLLGTISLSLCEEERSADEDDGEKEVKRGIFSLIKTAAKFVGKNLLKQAGKAGLEHLACKA
MLTLKKSMLLLFLLGTISLSLCEEERSADEDDGEKEVKRGIFSLIKTAAKFVGKNLLKQAGKAGLEHLACKA
NEC
GPVSAVLTCLRCTCLRVTLRVNPKTIGKLQVFPAGPQCSKVEVVASLKNGKQVCLDPEAPFLKKVIQKILDSGNKKN
MTKNHYKLYVDMKEGTHDYVSATVYSMFDEKVLWLPDIKKPENVADYTDEWKNGLYSYEKLAISELNDYNATFSL
LLAWCLVFLVIVQQTSSPVPQSDTPLTSVQEVQSRSLKRTARMTPLWRIMGTCPHGAYCQNHYECSGTGICRKGHC
SY
MINIPKMKFPKKYTEIISKKYKNKTPEEKAKIENDFIKDINDKDESEFYSPMMANMNEHELRAMLRMMPSLIDT
GDDNDD
MRSHSFLSALFLLVMMMPRGKAGVVPGEKQCILLKGVCCKDVSTSTDDTIGVCNDEKKCCRRWWVFDPYPT
PVPKKGKSP
MSTIVNYVCSMMIICLFQFTVATNFERKQISFSFFMKEYWPCVTDDDCPSDLCKKVDQIPKCVGGLCKCF
PIRFGQWER
MAKFASIIALLFAALVLFSAFEAPSMVEAQKLCEKSSGTWSGVCNNACKNQCNINLEGARHGSCNYIFPYH
RICICYFPC
MAKSAIIITFLFAALVLFSAFEAPIMVEAQKLCEKPSGTWSGVCNACKNQCNINLEGARHGSCNYIFPAH
KICICYFPC
MAPSRMVASAFLLAILVATEMGTTKVAEARHCLSQSHRFKGMCVSSNNCANVCRTESEFPDGECKSHGLER
KCFCKKVC
MNTLYHKMQEHVCFVGFTRDGVNYQTNRVRTFGEARHFITFNCSPSKDIRIFYRIPEHGAEDVEVVIARLDL
TETKENN
MTDNARKEYLNQFFGFKRYLYQDNERVAHIHVNGTYFHHGIVPGWQSVKKTFTDAEELEIYIKQHGLEEY
EEQKQLTLF
MAKTLKFFYTIILFSLFLVLQRKLTGCEVDGDCPKVFKLKMILFIKINNKCVRGLLSQTGTQCPDFFFLK
RTLPRFYF
MKQVCILLAVLLCTAAVADAMVFAYAPTARCKSIGARYCGYGLNRKGVSCDQTTINSCEDCKRKFGRCS
DGFITECFF
MTQILKLLVIVILFCFHICFVAEAEQVSDADCPKIFPGPRQHLLRCTQGNVCMVLVGQKNYFSIMSKTLF
SLLVIIIFLL
VSDMLEIFLIFGFGVYLFYRIAIIFLKSCKTIHTNIYEMLMLATIFMISTFAYKHQKTHILIAFLVMFMS
KLVQKGSYEE
FTFSILCQVNSDCLGEICLPPKTHWCNKILLEIYISCHLVMTLEPNLYLLPFLISWTRNNLYIILGLSLF
SRTNLSVLSWR
MAEIVKFVYLMIIIFLSTFLVSTKILEKHKCVTDGVEILEKGKCFDWEVCVRNSWLCVLDLVRCIKETCK
CIKILEPINVVPT
MFTNPKIDGETAEKDWFDSSEVEGEALFPLVDNKKSVRKYIFDSANMTNHGDGEGEEAFLKKILGEEYT
GNVTEGNEETL
MQRGKHMVEILEFVYAMILFLPLFLVITEVDGVDIYCETDADCPQITDWFYVVKCVDHKCELTKKLRL
LEYQTQKSAETPYI
MALSSQIWAACLLLLLLASLTSGSVFPOQTGQLAELQPDQDRAGARASWMPMFQRRRRRDTHFPICIFC
CGCHRSKCGMCKCT
MKLPVLFLFLFCFLDLLKTVKAEMKDTLFCFLKKGKCRHVCNMVEKRVGPCTKLNANCCIFVRDMRAI
IPEDQRTVSIKIRNKQN
MKTFSGAVTVAVVLVFCICIQSSASFPEVQEEAVSNDNAAAEHQETPVDSWMPNPNRQKRGFKCKFC
CGCCRAGVCGLCCKF
MKTISVAVTVAVVLVFCICIQSSASFPEAQEEAVSNDNAAAEHQETPVDSGMIPYNRQKRSFKCKFC
CGCCRAGVCGLCCKF
MKTSSVAVTVAVVLVFCICIQSSASFPEVQEEAVSNDNAAAEHQETSVDSWMPNPNRQKRSFKCKFC
CGCCRAGVCGLCCKF
MAKLMKLFYVMIHFISLLLITRNVRAYDDCYNHAECTNKIKCVPPRIAQCVRFKCDCIRLNNGPKTPW
SARPKRVHISPTRKNDF
MTIIIKFVNVLIIIFLSLFHVAKNDNDNKLSSFIEEGFLCFKDSDCPYNMCPSPLEKMCYFIKCVCGV
YGPPIRERRLYQSHNPMIQ
MYIVVYDVGVVERNVKVKFLRMHLNWNQNSVFEGEVTLAEFEIERIKEGLKKIIDENSDSVIIYKLRS
MPPRETGLIEKNPIEEII
MKSQNHAKFISFYKNDLFKIFQNNDSHFVFFALIIFLYTYLHVTNGVFVSCNSHIHCRVNNHKIGCN
IPEQYLLCVNLFCLWLDY
MNLAKGEEESLSDSLYAEALRCMCIKTTSGIHPKNIQSLEVIGKGTCHNQVEVIATLKDRKICLDPDAP
RIKKIVQKKLAGDESAD
MKYLIYATLVVYGLITLLTSYILTYGVYWLTNELSNMPMPWFIVLCGGIVLVITAFIILVNVNIQNK
IENQEVNIAIKEENDG
MAVFRVLLASLLISLLVLDVHADMVRCSSRPNLCHRACGTCCARCNCVAPGTSGNYDKCPCYGLSTH
GGRRKEVKEFSFFTHGS
MKTVLAFLFLTFAIFTYAESYEDVKEEIKNEVEKEIFEDLEESDALDSSVREFNDAKPWRFRRAIRRV
RWKRVAPYIPFVVKTVGKK
MKTVLAFLFLTFAIFTYAESYEDVKEEIKNEVEKEIFEDLEESDVLDSNVREFNDAKPWRFRRAIRRV
RWKRVAPYIPFVVRTVGKK
MKVAFVAVLLICLVLSSSLFEVSMAGSAFCSKCAKRCRAGMKDRCTRFCGICCSKCRVPSGTYGKHE
CPCYRDMKNSKGKPKCP
MAVMKSRVIVAAVLLAVVILSSLCPCYEAGGICGPKPTAPKRPCFSPYSEDHCDRQNCRFVCM
SHGYSDDGWCDEREVRKMCCCYH
MVRLVITYDIRKDKIRNKLFRLLERYGAWKQYSVFELEINPVHKVELFHSIADLIEDTDRVRIYDL
CERCQKGITELGEVSPDKMQVVI
RVPPYLGRDCKHWCARDNNQALYCCGPPGITYPPFIRKHGPKCPSVRSTCTGVRSSRPKFCPHDDA
CEFRSKCCYDACVKHHVCKTVEFY
MEFLKYPVSLVNHYAIGKSKRLVNRFDNILWYASEKIHGANASYALSATGEEYFAKRSGLISKDDKQ
FSMLPECVTSDIREGTTKAFRIL
S
MKAFSIAVAVTLVLAFCILQSSAIPVNGVKELEEAASNDTPVAARHEMSMQPWMLPNHIREKRQSHI
SMCTMCCNCCKNYKGC GFCCR
F
MKAFSIAVAVTLVLAFCIQCSSAVPFQGVQEELEAGGNDTPVAEHQVMSMESWMENPTRQKRHISHI
SLCRWCCNCCANKGCGFCCK
F
MKLVFATLLLCSSLLSSSFLEPVIAYEDSSYCSNKCSDRCSSAGVKDRCLRYCGICCAECKCVPSGT
YGNKHQPCYRDKLNKKGKPKC
P
MMMISLLVFNPEADGVVWVNYGQHASLLAKIDCGGACKARCRLSSRPHLCKRACGTCCQRCSCVPPG
TAGNYDVPCPYATLTTHGGKRC
CP
MNYEEFKKEIESIDYLSVENRYDRVLVYSSLGDYPLVAVSILETGFDVYNWRGHVRARDIPVLTEA
VEKFAETPYEERFPKLLPTYWR
KLC
MVVIVYVIVAYDVNVERVNRVKKFLRRYLNNWQNSLFEGELSSADLEEVKMGLREIINEDMVIYR
FRSADAFKREVMGIEKGLGGE
EVI
AISCQGVASAIAPCISYARGQSGSPSAGCCSGVKSLNNAARTTADRRAACNCLKNAAAGVSGLNAG
NAASIPSKCGVSIPTISTSTDC
SRVN

MKMFTVISYDIDVDDQRRTSVMKVLKGYGVRVQYSVFEAILDAREFHDLSNQLRKIIDPGQDSIRCYRLDQVAAQRTVIYIGIGLTTTDP
HYMV
MNSLYEQFKEGTLEKEGQRFVSVTGTIVLIDAEDPALPLKVIDIDGEGIKWLKEGAIKYMEPAAEKLYEVSIGGRLLSYYGSDYE
EGKWT
CF
FVEK
AVSNTVIADLYPCLSYVTQGGPVPTLCCNGLTTLKSQAQTSVDRQGVCRICKSAIGGLTSPRTIQNALELPSKCGVDLPYKFSP
STD
CDSIQ
MILQILGERKYDCKDSTEEPIVKPFIELIDGVQFLLEQENAFSLLNSKHIEITRVNPSPEYRTEKEGYKTFVNSVYVLNDEGKTL
RRL
FERGL
MKCILSLVTLFLVAVLVHSHPAEWNTHQQLDDALWEPAGEVTEEHVARLKRATCDLFSFRSKWTPNHAACAAHCLLRGNRGRCKGT
I
CHCRK
MSSRLAVFAYDIRDDRVRRLHALKTLREWRLDGQLSVHECQVDIAIQARRLEQLGDELDPATDAWLFTWVEGHRVLA
RGKGR
TTALQDG
LLAA
MTIDDLLVKFSLEKIDHNSDEYLLKQLLKMSYERIKNQCGVFELENLIGQELILIRARYAYQDLLEHFNDNYRPEIIDFSL
SLMEV
SE
DEESV
MVKFKVVREFKDIENQHKKYKVGELYPAEGYNNPRVELLTNQIKNKYDKVYIVPLDKLTKQELLELCESLQKKASSSMV
KSEI
IDLNG
EDND
GSGPTYCWNANPPGPNRCSNNKQCDGARTCSSSGFCQGTSRKPDGPKGPTYCWDEAKNPPGPNRCSNSKQCDGARTCSSSGFCQGT
AGHAAA
MANEIIKKTERFILVQIDKEGTERVLYQDFVGSFTTSDSASYAQDFKSEENAKKIAETLNLLYQLTVNQNGVKVVKEVVDRTDLSSDKS
VDSEIM
MEHLYIVSYDIRNQRRWRRLFKTMHGFGCWQLSVFQCRLDRIRIIKMEAAINEIVNHAEDHVLIDLGAENVKPKVSSIGKTFDPIL
RQAVIV
MKTFSVAVAVAVVLTFCICLQESSAVSFTEVQEQUEEPMSNDSPVAAHEEMSEESWKMPYNNRHKRSPAGRNKRRRRRCR
FCCGCCPNMIGC
GTCKF
MTVKHLIVCYDVEKTKDRNKVIKVLEYYGLIRVQYSVFMGSLTETRLHQMNAIKREFTKPSIKILVIEVCNACMERALLVHEELPKVN
RQFEVI
VTQYLVTTFKDSTGLPHEHITVARDNQFTTVVEAESKEEAKKEYEARNVPVDGATNLNDIKSNIGIFHVEKVEPNEGMVDINIETMKPF
EEADDD
MKAIPVALLLLVLVAAAASSFKHLAEAADGGAVPDGVC DGKCRSRLKAGRCMGLCMMCCGKCQGCVPSPGYASKDECPCYRDMKSPK
NQRPKCP
MANCIRRNALFFLTLLFLLSVSNLVQAARGGGKLPQQCNSKCSYRCSATSHKKPCMFFCLKCKKCLCVPPGTFGNKQTCPCYNNWKT
KEGRPKCP
MRIIALLAAILLVALQVRAGPLQARGDEAPGQEQRGPEDQDISISFAWDKSSALQVSGSTRGMVCSCRLVFCRRT
ELRVGNCLIGGVSF
TYCCTRVD
AQIVKLGDDGSLAFVPSKISVAAGEAIEFVNAGFPHNIVFDEDAVPAGVDADAISYDDYLNSKGETVVRKLSTPGVYGVYCEPHAGA
GMKMTITVQ
MKKLRTTTATTTLALILLVLIAATSLRVAMAGSAFCDSKCGVRCSKAGRHHDDCLKYCGICCAECNCVPSGTAGNKDECPCYRDKTTGH
GARTRPKCP
LDVPHVYL RDTKDCSKITIGNLPTEPFVTVSAYREPSDITVGMIPVSFIEGDFDKYYEDFASDITKRLKDMAKLKEDSHKQAVDSINK
EKVAEVMFFE
LNKKETLLNTTVTFLGGTSQSSRPKGKQTLCEMQKSDGSVEFIFDGTVFYVHSMVVVYMANKKHSMQATVSAIDGNYKYAGITFKLKVD
YKENTLLLEEI
MAQRFMFIYALIIFLSQFFVINTSMSIYITFKKFIIDFIHNVYHPSITSNFSLFNNAGDIPNNSNRNSPKEDVFCNSNDDCPTILYYV
SKCVYNFCEYW
MDIMDIEFLEEHLKQLVKEHVEQELKLMHPLKKLQVMTDWLGDTEKLSQGLDYFNDLTETELIEAMDASEIVESYSDVLLDFIDYYNI
DLTGLEEQLG